

Derivation and implementation of one-point quadrature quadrilateral shell element with MITC4+ method (MITC4+R) [☆]

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ARTICLE INFO

Keywords:

Degenerated shell
Quadrature element
The MITC4+ method
Shell structures
One point quadrature

ABSTRACT

In this work, a novel degenerated shell element called MITC4+R is developed, which is based on the MITC4+ (an improved MITC quadrilateral shell element) and reduced integration with one point quadrature. The assumed natural strain method is employed to construct the membrane and shear strains. A physical stabilization term is derived after the assumed natural strains are decomposed into the linear and a constant term with respect to the natural coordinates. The hourglass modes owing to one point quadrature is controlled in the computation of deformation by introducing this physical stabilization term. The developed degenerated shell element is successfully implemented into the implicit static and explicit dynamic analyses. In order to verify its computational accuracy and efficiency, several standard benchmarks and practical problems are performed. These results show that the MITC4+R element eliminates various locking problems common to shell elements, and it significantly improves the computational efficiency of the original MITC4+ element. Meanwhile, the MITC4+R owns the same accuracy and highly consistent convergence as the original MITC4+ element, making it more promising for practical applications.

1. Introduction

Nowadays, the simulation of shell elements has been a critical factor in the structural analysis of bodywork or airframe. In general, a mesh model of the body-in-white is established mainly from quadrilateral shell elements. The emergence of the shell element was made possible by the continuous development of plate elements. The first plate element was the Kirchhoff plate introduced by Adini and Clough [1]. In the following period, scholars continued refining Kirchhoff plate elements until 1969, when the Discrete Kirchhoff plate and shell elements appeared [2]. Unfortunately, Kirchhoff plates fail to accommodate transverse shear in the bending deformation. Thus, Raymond Mindlin's modification of the Kirchhoff plate so that it is middle normal remains straight after bending and deforming but is not necessarily perpendicular to the middle surface is called Mindlin plate element. Hughes and Tezduyar [3] first proposed a bilinear four-node quadrilateral shell element based on Mindlin's theory in 1977. Ahmad [4] applies the finite element formulation to propose a three-dimensional degenerate curved shell element; when the curvature of the shell element is zero, the

formula is the same as the Mindlin shell element formula. Scholars have developed some high-precision high-order quadrilateral degenerate shell elements on this basis [5–7]. Although higher-order elements usually perform better than lower-order elements in most situations, the calculation scale is much larger for the same number of meshes due to the higher number of nodes. It is a consistent goal of many researchers to build an efficient low-order four-node quadrilateral shell element without adding any nodes or degrees of freedom.

The main problems of low-order shell elements exist in various locking phenomenon, including shear locking, membrane locking, transverse shear locking, and volume locking. The shear locking is caused by the introduction of isoparametric elements, which can not accurately describe the in-plane bending deformation $W = x^2$ due to the linear interpolation of the low-order elements [8]. Membrane locking is caused by failure to satisfy the inextensibility constraint when the shell element bending deformation [9]. Transverse shear locking refers to the pseudo-shear phenomenon in the shell element during bending, making the deflection and bending strain much smaller than the expected results. Volume locking is spurious pressure stress occurring when the

[☆] This is the manuscript to be submitted to Computers & Structures.

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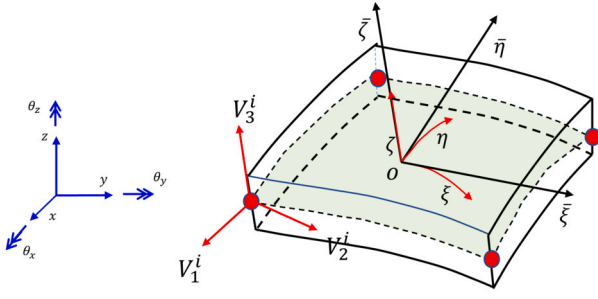


Fig. 1. A standard 4-node degenerated shell element.

Poisson's ratio is close to 0.5. Many methods have been introduced to avoid various locking. For example, the reduced/selective integration method [10,11] introduced between 1977 and 1978, the assumed natural strain method [12] introduced between 1981 and 1982; and the subsequent discrete shear gap method [13,14] and the enhanced strain method [15,16], etc. Many scholars have developed elements that have eliminated various locking phenomena, Katili et al. [17] developed a 24 dof quadrilateral shell element called DKMQ24 based on the DKMQ plate element without shear locking and membrane locking phenomena. Selective reduction of the integral of the shear term was employed in the DKMQ24 called DKMQ24+ was introduced by Štembera and Füssl [18]. The mixed interpolation of tensorial components (MITC) method [19–21] is similar to a particular assumed strain method. It effectively eliminates membrane locking and shear locking in applying shell elements, especially in high-order shell elements [22–25]. However, membrane locking still exists when the 4-node quadrilateral MITC4 shell element is calculated for some cured models. Although researchers have made many attempts to modify the strain components of the membrane [26,27], these attempts have not shown better results and have even demonstrated worse results. Then, the MITC4+ method is introduced by Yeongbin et al. [28], which constructs a new assumed strain by four non-overlapping triangular strain fields in the mid-surface of the quadrilateral shell element. This method provides an excellent solution to the membrane-locking problem of MITC4 elements, and the convergence of MITC4+ elements was verified by Ko et al. [29], through a set of widely-used benchmark problems. It shows well the performance of MITC4+ elements even under distorted meshes. However, the MITC4+ element computational scale and the computational efficiency are still not up to the ideal conditions.

In the past few decades, researchers have been keen to develop reduced integration methods to improve the computational efficiency of finite elements. Currently, the most dominant reduced integration method for shell element is the one-point integration (at the centre of a mid-surface). It is widely used in commercial software for the simulation of collisions and ramming, such as LS-DYNA and RADIOSS, because of its superior computational efficiency for static and explicit dynamics problems. However, zero energy and hourglass models will be induced in the centre point of the integral sometimes. For restraining the hourglass model, various hourglass control methods were published. Kosloff and Frazier [30] first attempted hourglass control in a two-dimensional quadrilateral, this method was applied to the shell element by Taylor [31], Hughes and Liu [32]. The introduction of the BT shell [33] was a milestone in developing the one-point integral shell element, which was popular because of the better stability obtained by adding artificial strain. However, the BT shell had poor results in warping behaviour and required manual adjustment of the hourglass control factor. Subsequently, Belytschko and Leviathan [34] proposed a one-point integral four-node shell element based on physical stabilization, sacrificing a smaller computational efficiency to obtain better stability. Many researchers [35–37] have used this method to derive a series of shell elements.

In this work, a novel one point quadrature quadrilateral shell element with the MITC4+ method is developed to improve the compu-

tational efficiency of the MITC4+ shell element. The key idea of this work is to construct a stabilization part in quadrilateral shell element with one point quadrature. The strain in the shell element is divided into two components, the membrane strain field and the bending strain field, in the natural coordinate system. A physical stabilization scheme is used to decompose these two components of the strain field into the sum of a constant and a sum of linear terms, respectively. In this way, the developed MITC4+R element can alleviate the hourglass phenomenon, which makes the MITC4+R to be more efficient than the original MITC4+ without any loss of accuracy. Standard benchmarks and practical examples are performed. The results are compared with analytical or previously reported results. The performance of the developed element has great potential for practical applications.

2. Theoretical basics

2.1. Unified derivation of degenerated shell element

In this section, we briefly review the formulation of the continuum-based degenerated shell formulation. The current position of each point on the shell can be defined by

$$\mathbf{x}(\xi, \eta, \zeta) = \bar{\mathbf{x}} + \frac{\zeta}{2} \mathbf{V}_3, \quad (1)$$

where $\bar{\mathbf{x}}$ is the position vector of mid-surface.

$$\mathbf{x}(\xi, \eta, \zeta) = \sum_{i=1}^n N_i(\xi, \eta) \mathbf{x}_i + \frac{\zeta}{2} \sum_{i=1}^n a_i N_i(\xi, \eta) \mathbf{V}_n^i, \quad (2)$$

where n is the number of the nodes within one shell element; $N_i(\xi, \eta)$ is the two-dimensional interpolation function of the standard isoparametric procedure corresponding to node i ; $\mathbf{x}_i$ is the position vector of node i in the global Cartesian coordinate system, and a_i and $\mathbf{V}_n^i$ is the shell thickness and director vector at node i , see Fig. 1.

The corresponding displacement interpolation of the element is,

$$\mathbf{u}(\xi, \eta, \zeta) = \sum_{i=1}^n N_i(\xi, \eta) \mathbf{u}_i + \frac{\zeta}{2} \sum_{i=1}^n a_i N_i(\xi, \eta) (-\mathbf{V}_2^i \theta_{xi} + \mathbf{V}_1^i \theta_{yi}), \quad (3)$$

in which $\mathbf{u}_i$ is the nodal displacement vector in global Cartesian coordinate system; $\mathbf{V}_1^i$ and $\mathbf{V}_2^i$ are unit vector orthogonal to $\mathbf{V}_n^i$ and to each other; θ_{xi} and θ_{yi} are the rotations of the director vector $\mathbf{V}_n^i$ about unit vector $\mathbf{V}_1^i$ and $\mathbf{V}_2^i$ at node i .

The linear terms of the covariant strain components $\bar{\epsilon}_{ij}$ in the convective coordinate system are given by

$$\bar{\epsilon}_{ij} = \frac{1}{2} (\mathbf{g}_i \cdot \mathbf{u}_j + \mathbf{g}_j \cdot \mathbf{u}_i) \quad (i, j = 1, 2, 3) \quad (4)$$

in which,

$$\mathbf{g}_i = \frac{\partial \mathbf{x}}{\partial r_i}, \quad \mathbf{u}_i = \frac{\partial \mathbf{u}}{\partial r_i} \quad (r_1 = \xi, r_2 = \eta, r_3 = \zeta) \quad (5)$$

The covariant strains can be obtained directly using Eq. (4), and the covariant strain components can be easily transformed into the global Cartesian coordinate system by

$$\epsilon = \bar{\epsilon}_{ij} (\mathbf{g}^i \otimes \mathbf{g}^j) \quad (6)$$

where $\mathbf{g}^i = \partial \mathbf{x} / \partial r_i$ ($r_i = \xi, \eta, \zeta$). For convenience, all strain components in the following sections are the covariant strain components unless otherwise stated. In this work, we divided the strain of the shell into two components

$$\bar{\epsilon}_{ij} = \bar{\epsilon}_{ij}^m + \bar{\epsilon}_{ij}^{bs} \quad (7)$$

where $\bar{\epsilon}_{ij}^m$ is the membrane strain, and the remaining component $\bar{\epsilon}_{ij}^{bs}$ is transverse shear and bending strain, separately. The membrane strain is given

$$\bar{\epsilon}_{ij}^m = \frac{1}{2} \left(\frac{\partial \mathbf{x}_m}{\partial r_i} \cdot \frac{\partial \mathbf{u}_m}{\partial r_j} + \frac{\partial \mathbf{x}_m}{\partial r_j} \cdot \frac{\partial \mathbf{u}_m}{\partial r_i} \right) \quad (i, j = 1, 2) \quad (8)$$

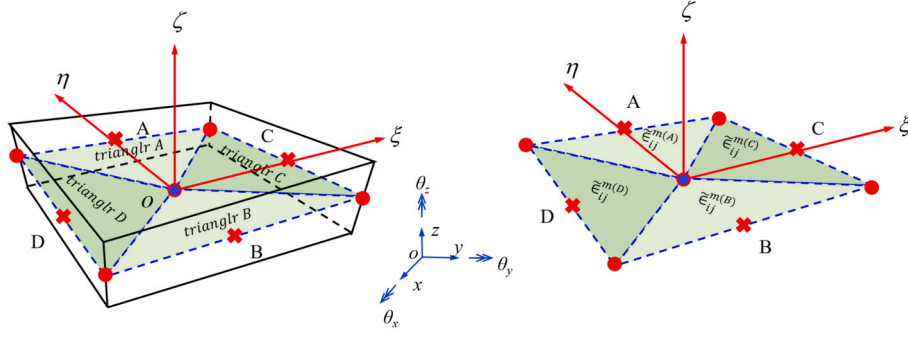


Fig. 2. The MITC4+ membrane strain field.

$\bar{\epsilon}_{ij}^{bs}$ is composed of in-plane due to bending $\bar{\epsilon}_{ij}^b$, thickness strain $\bar{\epsilon}_{33}$ and the transverse shear strain $\bar{\epsilon}_{ij}^s$.

$$\bar{\epsilon}_{ij}^{bs} = \bar{\epsilon}_{ij}^b + \bar{\epsilon}_{33} + \bar{\epsilon}_{ij}^s. \quad (9)$$

In-plane strain due to bending is given

$$\bar{\epsilon}_{ij}^b = \zeta \bar{\epsilon}_{ij}^{b1} + \zeta^2 \bar{\epsilon}_{ij}^{b2} \quad (i, j = 1, 2) \quad (10)$$

with

$$\begin{aligned} \bar{\epsilon}_{ij}^{b1} &= \frac{1}{2} \left(\frac{\partial \mathbf{x}_m}{\partial r_i} \cdot \frac{\partial \mathbf{u}_b}{\partial r_j} + \frac{\partial \mathbf{x}_m}{\partial r_j} \cdot \frac{\partial \mathbf{u}_b}{\partial r_i} + \frac{\partial \mathbf{x}_b}{\partial r_i} \cdot \frac{\partial \mathbf{u}_m}{\partial r_j} + \frac{\partial \mathbf{x}_b}{\partial r_j} \cdot \frac{\partial \mathbf{u}_m}{\partial r_i} \right) \\ \bar{\epsilon}_{ij}^{b2} &= \frac{1}{2} \left(\frac{\partial \mathbf{x}_b}{\partial r_i} \cdot \frac{\partial \mathbf{u}_b}{\partial r_j} + \frac{\partial \mathbf{x}_b}{\partial r_j} \cdot \frac{\partial \mathbf{u}_b}{\partial r_i} \right) \end{aligned} \quad (11)$$

The corresponding geometry and displacement interpolation functions are as follows

$$\begin{aligned} \mathbf{x}_m &= \sum_{i=1}^4 N_i(\xi, \eta) \mathbf{x}_i, \quad \mathbf{x}_b = \frac{1}{2} \sum_{i=1}^4 a_i N_i(\xi, \eta) \mathbf{V}_n^i, \\ \mathbf{u}_m &= \sum_{i=1}^4 h_i(\xi, \eta) \mathbf{u}_i, \quad \mathbf{u}_b = \frac{1}{2} \sum_{i=1}^4 a_i N_i(\xi, \eta) (-\mathbf{V}_2^i \alpha_i + \mathbf{V}_1^i \beta_i) \end{aligned} \quad (12)$$

To satisfy the shell statics assumption ($\bar{\sigma}_{33} = 0$), the thickness strain ($\bar{\epsilon}_{33}$) is calculated independently using the plane stress constitutive equation.

$$\bar{\epsilon}_{33} = \frac{1}{2} \left(\frac{\partial \mathbf{x}}{\partial \zeta} \cdot \frac{\partial \mathbf{u}}{\partial \zeta} + \frac{\partial \mathbf{x}}{\partial \zeta} \cdot \frac{\partial \mathbf{u}}{\partial \zeta} \right) \quad (13)$$

The shear strain $\bar{\epsilon}_{ij}^s$ should be a separate interpolation in this work, which is discussed in Section 2.3.

2.2. The MITC4+ method

The key idea of the MITC4 method is to construct a new assumed strain field in the mid-surface to eliminate various locking phenomenon of the shell element. The new assumed strain field is interpolated by sampling points (A, B, C, D) located at the centre of each edge of the quadrilateral shell with natural coordinates (0, 1, 0), (0, -1, 0), (1, 0, 0) and (-1, 0, 0), respectively.

The MITC4+ method was developed by extending the MITC method to overcome membrane locking by constructing an assumed membrane strain field. The assumed membrane strain field was derived from four non-overlapping 3-node triangular domains divided by point O and four nodes of quadrilateral shell as shown in Fig. 2. Point O is defined as the centroid of the mid-surface in the shell element, which is calculated from the four nodes of the quadrilateral shell, and the detailed derivation is shown in [28]. The strain in each triangular domain is given by

$$\bar{\epsilon}_{ij}^m = \frac{1}{2} \left(\frac{\partial \hat{\mathbf{x}}}{\partial r_i} \cdot \frac{\partial \hat{\mathbf{u}}}{\partial r_j} + \frac{\partial \hat{\mathbf{x}}}{\partial r_j} \cdot \frac{\partial \hat{\mathbf{u}}}{\partial r_i} \right), \quad (i, j = 1, 2) \quad (14)$$

While $\bar{\epsilon}_{ij}^m$ is the strain in the natural coordinate system of each triangular domain and the strains are constant in each triangular domain, the strain field at each sampling point can be treated as $\bar{\epsilon}_{ij}^m$ corresponding to the triangular domain, as follows:

$$\begin{aligned} \bar{\epsilon}_{ij}^m(0, 1, 0) &= \bar{\epsilon}_{ij}^{m(A)} (\mathbf{g}_i \cdot \hat{\mathbf{g}}^k) (\mathbf{g}_j \cdot \hat{\mathbf{g}}^l), & \bar{\epsilon}_{ij}^m(0, -1, 0) &= \bar{\epsilon}_{ij}^{m(B)} (\mathbf{g}_i \cdot \hat{\mathbf{g}}^k) (\mathbf{g}_j \cdot \hat{\mathbf{g}}^l) \\ \bar{\epsilon}_{ij}^m(1, 0, 0) &= \bar{\epsilon}_{ij}^{m(C)} (\mathbf{g}_i \cdot \hat{\mathbf{g}}^k) (\mathbf{g}_j \cdot \hat{\mathbf{g}}^l), & \bar{\epsilon}_{ij}^m(-1, 0, 0) &= \bar{\epsilon}_{ij}^{m(D)} (\mathbf{g}_i \cdot \hat{\mathbf{g}}^k) (\mathbf{g}_j \cdot \hat{\mathbf{g}}^l) \end{aligned} \quad (15)$$

Where, $\bar{\epsilon}_{ij}^m$ is the constant strain in natural coordinates, $\hat{\mathbf{g}}$ is the covariant base vectors.

$$\hat{\mathbf{g}}_i = \frac{\partial \hat{\mathbf{x}}}{\partial r_i}, \quad (i = 1, 2) \quad (16)$$

and

$$\hat{\mathbf{g}}_3 = \mathbf{g}_3 \quad (17)$$

The new assumed membrane strain field of the shell element can be interpolated by

$$\bar{\epsilon}_{ij}^m = a + b\xi + c\eta + d\xi\eta, \quad (i, j = 1, 2) \quad (18)$$

The a , b , c and d of Eq. (18) are given as

$$\begin{aligned} a &= \frac{1}{4} (\bar{\epsilon}_{ij}^{m(A)} + \bar{\epsilon}_{ij}^{m(B)} + \bar{\epsilon}_{ij}^{m(C)} + \bar{\epsilon}_{ij}^{m(D)}), \\ b &= \frac{1}{2} (-\bar{\epsilon}_{ij}^{m(D)} + \bar{\epsilon}_{ij}^{m(C)}), \\ c &= \frac{1}{2} (-\bar{\epsilon}_{ij}^{m(B)} + \bar{\epsilon}_{ij}^{m(A)}), \\ d &= 0 \end{aligned} \quad (19)$$

Therefore, the MITC4+ shell element membrane strain can be obtained

$$\begin{aligned} \bar{\epsilon}_{ij}^m &= \frac{1}{4} (\bar{\epsilon}_{ij}^{m(A)} + \bar{\epsilon}_{ij}^{m(B)} + \bar{\epsilon}_{ij}^{m(C)} + \bar{\epsilon}_{ij}^{m(D)}) + \frac{1}{2} (-\bar{\epsilon}_{ij}^{m(D)} + \bar{\epsilon}_{ij}^{m(C)}) \xi \\ &\quad + \frac{1}{2} (-\bar{\epsilon}_{ij}^{m(B)} + \bar{\epsilon}_{ij}^{m(A)}) \eta \end{aligned} \quad (20)$$

2.3. Derivation of strain tensor

The quadrilateral displacement field can be considered as composed of the following basic displacement modes: rigid body term, linear term and bilinear term with $\xi\eta$, where the deformation reflected by the bilinear term displacement contains the hourglass mode of the quadrilateral such as Fig. 3, and these bilinear term displacement fields correspond to the linear term part of the strain. To restrain the hourglass phenomenon, the linear term strain requires particular treatment.

The degenerated shell strain field can be expressed as a displacement and rotation-related term.

$$\bar{\epsilon} = \{ \bar{\epsilon}_{11} \quad \bar{\epsilon}_{22} \quad \bar{\epsilon}_{33} \quad \bar{\epsilon}_{12} \quad \bar{\epsilon}_{13} \quad \bar{\epsilon}_{23} \}^T = \bar{\mathbf{B}}_i \mathbf{d}_i \quad (i = 1, 2, 3, 4) \quad (21)$$

with

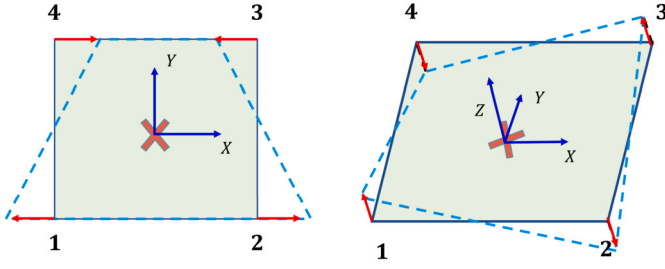


Fig. 3. The hourglass mode of the quadrilateral.

$$\mathbf{d}_i = \{u_i \quad v_i \quad w_i \quad \theta_{xi} \quad \theta_{yi}\}^T \quad (22)$$

According to Eq. (7) and Eq. (9), Eq. (21) can be written as

$$\begin{aligned} \bar{\epsilon} &= \{\bar{\epsilon}_{11}^m + \bar{\epsilon}_{11}^b \quad \bar{\epsilon}_{22}^m + \bar{\epsilon}_{22}^b \quad \bar{\epsilon}_{33} \quad \bar{\epsilon}_{12}^m + \bar{\epsilon}_{12}^b \quad \bar{\epsilon}_{13}^s \quad \bar{\epsilon}_{23}^s\}^T \\ &= \bar{\mathbf{B}}_i \mathbf{d}_i \quad (i = 1, 2, 3, 4) \end{aligned} \quad (23)$$

where membrane strain is obtained as

$$\bar{\epsilon}^m = \{\bar{\epsilon}_{11}^m \quad \bar{\epsilon}_{22}^m \quad 0 \quad \bar{\epsilon}_{12}^m \quad 0 \quad 0\}^T = \bar{\mathbf{B}}_i^m \mathbf{d}_i \quad (i = 1, 2, 3, 4) \quad (24)$$

and the remaining strain caused by transverse shear and bending can be expressed as

$$\bar{\epsilon}^{bs} = \{\bar{\epsilon}_{11}^b \quad \bar{\epsilon}_{22}^b \quad \bar{\epsilon}_{33} \quad \bar{\epsilon}_{12}^b \quad \bar{\epsilon}_{13}^s \quad \bar{\epsilon}_{23}^s\}^T = \bar{\mathbf{B}}_i^{bs} \mathbf{d}_i \quad (25)$$

The membrane strain of MITC4+R elements is replaced by Eq. (20) for the conventional shell element membrane strain. The final form of membrane strain field is divided into constant field $\bar{\epsilon}^{m0}$ and linear field $\bar{\epsilon}^{mH}$ in ξ and η .

$$\bar{\epsilon}^m = \bar{\epsilon}^{m0} + \bar{\epsilon}^{mH} \quad (26)$$

the MITC4+R element membrane strain can be rewritten

$$\begin{aligned} \bar{\epsilon}_{ij}^{m0} &= \frac{1}{4}(\bar{\epsilon}_{ij}^{m(A)} + \bar{\epsilon}_{ij}^{m(B)} + \bar{\epsilon}_{ij}^{m(C)} + \bar{\epsilon}_{ij}^{m(D)}), \\ \bar{\epsilon}_{ij}^{mH} &= \frac{1}{2}(-\bar{\epsilon}_{ij}^{m(D)} + \bar{\epsilon}_{ij}^{m(C)})\xi + \frac{1}{2}(-\bar{\epsilon}_{ij}^{m(B)} + \bar{\epsilon}_{ij}^{m(A)})\eta \end{aligned} \quad (27)$$

Then, Eq. (21) is substituted into Eq. (15). The strain in each triangular domain can be obtained as

$$\begin{aligned} \bar{\epsilon}_{ij}^{m(A)} &= \bar{\mathbf{B}}_A^m \mathbf{d}_i, \quad \bar{\epsilon}_{ij}^{m(B)} = \bar{\mathbf{B}}_B^m \mathbf{d}_i, \\ \bar{\epsilon}_{ij}^{m(C)} &= \bar{\mathbf{B}}_C^m \mathbf{d}_i, \quad \bar{\epsilon}_{ij}^{m(D)} = \bar{\mathbf{B}}_D^m \mathbf{d}_i. \end{aligned} \quad (28)$$

Combined with Eq. (27) - Eq. (28). The B matrix of the constant and linear term can be constructed, respectively.

$$\bar{\epsilon}^{m0} = \bar{\mathbf{B}}_{1i}^{m0} \mathbf{d}_i, \quad \bar{\epsilon}^{mH} = (\xi \bar{\mathbf{B}}_{1i}^{mH} + \eta \bar{\mathbf{B}}_{2i}^{mH}) \mathbf{d}_i \quad (29)$$

and

$$\bar{\mathbf{B}}_{1i}^{m0} = \frac{1}{4}(\bar{\mathbf{B}}_A^m + \bar{\mathbf{B}}_B^m + \bar{\mathbf{B}}_C^m + \bar{\mathbf{B}}_D^m) \quad (30)$$

with

$$\begin{aligned} \bar{\mathbf{B}}_{1i}^{mH} &= \frac{1}{2}(-\bar{\mathbf{B}}_D^m + \bar{\mathbf{B}}_C^m) \\ \bar{\mathbf{B}}_{2i}^{mH} &= \frac{1}{2}(-\bar{\mathbf{B}}_B^m + \bar{\mathbf{B}}_A^m) \end{aligned} \quad (31)$$

Similarly, the transverse shear and bending strain can be decomposed into constant term $\bar{\epsilon}_{ij}^{bs0}$ and linear terms $\bar{\epsilon}_{ij}^{bsH}$

$$\bar{\epsilon}_{ij}^{bs} = \bar{\epsilon}_{ij}^{bs0} + \bar{\epsilon}_{ij}^{bsH} \quad (32)$$

In particular, the in-plane due to bending strain expressions are shown in Eq. (11), expressed in terms of the Jacobian as:

$$\begin{aligned} \bar{\epsilon}_{ij}^{b1} &= \frac{1}{2}(\mathbf{J}_j^m \cdot \frac{\partial \mathbf{u}_b}{\partial r_j} + \mathbf{J}_i^m \cdot \frac{\partial \mathbf{u}_b}{\partial r_i} + \mathbf{J}_j^b \cdot \frac{\partial \mathbf{u}_m}{\partial r_j} + \mathbf{J}_i^b \cdot \frac{\partial \mathbf{u}_m}{\partial r_i}) \\ \bar{\epsilon}_{ij}^{b2} &= \frac{1}{2}(\mathbf{J}_j^b \cdot \frac{\partial \mathbf{u}_b}{\partial r_j} + \mathbf{J}_i^b \cdot \frac{\partial \mathbf{u}_b}{\partial r_i}) \end{aligned} \quad (i, j = 1, 2) \quad (33)$$

$\mathbf{J}^m$ is the coordinate conversion relationship of the Jacobian of the membrane component, and $\mathbf{J}^b$ is the bending component. Similar to the strain field, the Jacobian matrix can be decomposed into

$$\mathbf{J}^m = \frac{1}{4} \begin{bmatrix} \xi_i x_i & \xi_i y_i & \xi_i z_i \\ \eta_i x_i & \eta_i y_i & \eta_i z_i \\ 0 & 0 & 0 \end{bmatrix} + \frac{1}{4} \begin{bmatrix} h_i \eta x_i & h_i \eta y_i & h_i \eta z_i \\ h_i \xi x_i & h_i \xi y_i & h_i \xi z_i \\ 0 & 0 & 0 \end{bmatrix} \quad (34)$$

and

$$\mathbf{J}^b = \frac{a}{8} \begin{bmatrix} \xi_i V_{n_x}^i & \xi_i V_{n_y}^i & \xi_i V_{n_z}^i \\ \eta_i V_{n_x}^i & \eta_i V_{n_y}^i & \eta_i V_{n_z}^i \\ 0 & 0 & 0 \end{bmatrix} + \frac{a}{8} \begin{bmatrix} h_i \eta V_{n_x}^i & h_i \eta V_{n_y}^i & h_i \eta V_{n_z}^i \\ h_i \xi V_{n_x}^i & h_i \xi V_{n_y}^i & h_i \xi V_{n_z}^i \\ 0 & 0 & 0 \end{bmatrix} \quad (35)$$

Then, the thickness strain can be given as

$$\bar{\epsilon}_{33} = \frac{1}{2}(\mathbf{J}_3 \cdot \frac{\partial \mathbf{u}}{\partial \zeta} + \mathbf{J}_3 \cdot \frac{\partial \mathbf{u}}{\partial \zeta}) \quad (36)$$

with the Jacobian at the centroid point of the mid-surface

$$\mathbf{J} = \frac{1}{4} \begin{bmatrix} \xi_i x_i & \xi_i y_i & \xi_i z_i \\ \eta_i x_i & \eta_i y_i & \eta_i z_i \\ 0.5aV_{n_x}^i & 0.5aV_{n_y}^i & 0.5aV_{n_z}^i \end{bmatrix} + \frac{1}{4} \begin{bmatrix} h_i \eta x_i & h_i \eta y_i & h_i \eta z_i \\ h_i \xi x_i & h_i \xi y_i & h_i \xi z_i \\ 0 & 0 & 0 \end{bmatrix} \quad (37)$$

Many shell elements suffer from transversal shear locking; independent correction is necessary for transverse shear strain. The MITC4+ follows the MITC4 method of directly interpolating the assumed transverse shear strain.

$$\begin{aligned} \Delta \bar{\epsilon}_{13}^s &= \frac{1}{2}(1 + \eta)\Delta \bar{\epsilon}_{13}^A + \frac{1}{2}(1 - \eta)\Delta \bar{\epsilon}_{13}^C \\ \Delta \bar{\epsilon}_{23}^s &= \frac{1}{2}(1 + \xi)\Delta \bar{\epsilon}_{23}^D + \frac{1}{2}(1 - \xi)\Delta \bar{\epsilon}_{23}^B \end{aligned} \quad (38)$$

The same sampling node were used as the MITC4+ method to interpolate the assumed strain in the transverse shear direction.

$$\begin{aligned} \Delta \bar{\epsilon}_{13}^A &= \frac{1}{4}(\xi_i + h_i)\mathbf{J}_3^A \cdot \mathbf{d}_i - \frac{a_i}{8}(1 + \eta_i)\{B_{11i}^A \quad B_{21i}^A\} \theta_i, \\ \Delta \bar{\epsilon}_{23}^B &= \frac{1}{4}(\eta_i - h_i)\mathbf{J}_3^B \cdot \mathbf{d}_i - \frac{a_i}{8}(1 - \xi_i)\{B_{12i}^B \quad B_{22i}^B\} \theta_i, \\ \Delta \bar{\epsilon}_{13}^C &= \frac{1}{4}(\xi_i - h_i)\mathbf{J}_3^C \cdot \mathbf{d}_i - \frac{a_i}{8}(1 - \eta_i)\{B_{11i}^C \quad B_{21i}^C\} \theta_i, \\ \Delta \bar{\epsilon}_{23}^D &= \frac{1}{4}(\eta_i + h_i)\mathbf{J}_3^D \cdot \mathbf{d}_i - \frac{a_i}{8}(1 + \eta_i)\{B_{12i}^D \quad B_{22i}^D\} \theta_i. \end{aligned} \quad (39)$$

According to Eq. (25) and Eq. (33) - Eq. (39), the constants without $\zeta^2 \bar{\mathbf{B}}_{3i}^{bs0}$ and linear terms without $\xi \eta \bar{\mathbf{B}}_{3i}^{bsH}$ are expressed as follows:

$$\begin{aligned} \bar{\epsilon}^{bs0} &= (\bar{\mathbf{B}}_{1i}^{bs0} + \zeta \bar{\mathbf{B}}_{2i}^{bs0}) \mathbf{d}_i, \\ \bar{\epsilon}^{bsH} &= (\xi \bar{\mathbf{B}}_{1i}^{bsH} + \eta \bar{\mathbf{B}}_{2i}^{bsH} + \xi \zeta \bar{\mathbf{B}}_{4i}^{bsH} + \eta \zeta \bar{\mathbf{B}}_{5i}^{bsH}) \mathbf{d}_i \end{aligned} \quad (40)$$

The detailed expression of Eq. (40) is shown in Appendix A

3. Implementation of finite element formulation

3.1. Formulation of the linear stiffness matrix

To satisfy the stress-strain law, all simulations need to be transformed to co-rotating coordinate system by transformation matrix $\mathbf{T}$, which is given in Ref [38].

$$\bar{\epsilon} = \mathbf{T} \bar{\epsilon} \quad (41)$$

According to the previous derivation, the total stiffness matrix is decomposed into a constant term and a linear term in ξ and η

$$\mathbf{K}_{ij} = \mathbf{K}_{ij}^0 + \mathbf{K}_{ij}^{stab}. \quad (42)$$

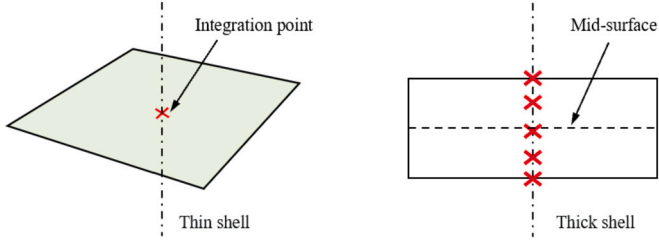


Fig. 4. Integration point in the direction of shell thickness.

$\mathbf{K}_{ij}^0$ is evaluated directly at the centre of the mid-surface ($\xi, \eta, \zeta = 0$). When the shell is thick, numerical integration of ζ can be performed in the thickness direction, such as an odd number of integration points in the thickness direction of the shell for Gaussian integration shown in Fig. 4

$$\mathbf{K}_{ij}^0 = \int_{-1}^1 \int_{-1}^1 \int_{-1}^1 (\tilde{\mathbf{B}}^0)^T \mathbf{D}(\tilde{\mathbf{B}}^0) d\xi d\eta d\zeta \quad (43)$$

and

$$\tilde{\mathbf{B}}^0 = \tilde{\mathbf{B}}_{1i}^{m0} + \tilde{\mathbf{B}}_{1i}^{bs0} + \zeta \tilde{\mathbf{B}}_{2i}^{bs0} \quad (44)$$

The physical stabilization part uses simple integration:

$$\mathbf{K}_{ij}^{stab} = \int_{-1}^1 \int_{-1}^1 \int_{-1}^1 (\tilde{\mathbf{B}}^H)^T \mathbf{D}(\tilde{\mathbf{B}}^H) d\xi d\eta d\zeta \quad (45)$$

with

$$\tilde{\mathbf{B}}^H = \xi(\tilde{\mathbf{B}}_{1i}^{mH} + \tilde{\mathbf{B}}_{1i}^{bH}) + \eta(\tilde{\mathbf{B}}_{2i}^{mH} + \tilde{\mathbf{B}}_{2i}^{bH}) + \xi\zeta\tilde{\mathbf{B}}_{4i}^{bH} + \eta\zeta\tilde{\mathbf{B}}_{5i}^{bH} \quad (46)$$

Eq. (45) can be rewritten as:

$$\begin{aligned} \mathbf{K}_{ij}^{stab} = & \int_{-1}^1 \int_{-1}^1 \int_{-1}^1 (\tilde{\mathbf{B}}_{1i}^H)^T \mathbf{D}(\tilde{\mathbf{B}}_{1i}^H) \xi^2 \det J_0 d\xi d\eta d\zeta \\ & + \int_{-1}^1 \int_{-1}^1 \int_{-1}^1 (\tilde{\mathbf{B}}_{2i}^H)^T \mathbf{D}(\tilde{\mathbf{B}}_{2i}^H) \eta^2 \det J_0 d\xi d\eta d\zeta \\ & + \int_{-1}^1 \int_{-1}^1 \int_{-1}^1 (\tilde{\mathbf{B}}_{4i}^H)^T \mathbf{D}(\tilde{\mathbf{B}}_{4i}^H) \xi^2 \zeta^2 \det J_0 d\xi d\eta d\zeta \\ & + \int_{-1}^1 \int_{-1}^1 \int_{-1}^1 (\tilde{\mathbf{B}}_{5i}^H)^T \mathbf{D}(\tilde{\mathbf{B}}_{5i}^H) \eta^2 \zeta^2 \det J_0 d\xi d\eta d\zeta. \end{aligned} \quad (47)$$

The simply integration method in the above equation is as follows:

$$\begin{aligned} \int_{-1}^1 \int_{-1}^1 \int_{-1}^1 \xi^2 d\xi d\eta d\zeta &= \frac{8}{3}, \\ \int_{-1}^1 \int_{-1}^1 \int_{-1}^1 \eta^2 d\xi d\eta d\zeta &= \frac{8}{3}, \\ \int_{-1}^1 \int_{-1}^1 \int_{-1}^1 \xi^2 \zeta^2 d\xi d\eta d\zeta &= \frac{8}{9}, \\ \int_{-1}^1 \int_{-1}^1 \int_{-1}^1 \eta^2 \zeta^2 d\xi d\eta d\zeta &= \frac{8}{9}. \end{aligned} \quad (48)$$

3.2. Internal force vector

The critical of the explicit algorithm is the solution to the internal forces. It is for dynamic large deformation problems, the energy function in the updated Lagrangian formulation can be written as

$$\delta W^{ext} + \delta W^{kin} + \delta W^{con} = \delta W^{int} \quad (49)$$

where δW^{ext} , δW^{kin} , δW^{con} and δW^{int} are the external energy, the kinetic energy, the contact friction energy and the strain energy. Their expression can be written as

$$\begin{aligned} \delta W^{ext} &= \int_{\Omega} \delta \mathbf{u} \cdot \mathbf{b} d\Omega + \int_{\Gamma_t} \delta \mathbf{u} \cdot \mathbf{t} d\Gamma \\ \delta W^{kin} &= \int_{\Omega} \rho \delta \mathbf{u} : \ddot{\mathbf{u}} d\Omega \\ \delta W^{con} &= \int_{\Gamma_c} (q_N \delta \gamma_N + q_T \delta \gamma_T) d\Gamma \\ \delta W^{int} &= \int_{\Omega} \delta \epsilon : \tau d\Omega \end{aligned} \quad (50)$$

where q_N and q_T represent the contact force and friction force, γ_N and γ_T represent the relative displacements of the tangential and normal directions at the contact interface Γ_c . The discrete motion equation can be written as follows

$$\mathbf{m}_i \ddot{\mathbf{u}}_i = \mathbf{F}_i^{con} + \mathbf{F}_i^{ext} - \mathbf{F}_i^{int} \quad (51)$$

Where $\mathbf{m}_i$ is lump mass matrix, $\ddot{\mathbf{u}}_i$ is acceleration and $\mathbf{F}_i^{con}$, $\mathbf{F}_i^{ext}$, $\mathbf{F}_i^{int}$ represent external, contact and internal forces at node i . The internal force vector is defined by

$$\mathbf{F}_i^{int} = \int_{\Omega} \tilde{\mathbf{B}}_i^T \tilde{\sigma} d\Omega \quad (52)$$

Similar to strain, the internal force vector be decomposed into a constant term and a linear term

$$\mathbf{F}_i^{int} = \mathbf{F}_i^0 + \mathbf{F}_i^{stab}. \quad (53)$$

where

$$\mathbf{F}_i^0 = \int_{\Omega} (\tilde{\mathbf{B}}_i^0)^T \tilde{\sigma}_{n+1}^0 d\Omega, \quad (54)$$

$$\mathbf{F}_i^{stab} = \int_{-1}^1 \int_{-1}^1 \int_{-1}^1 (\tilde{\mathbf{B}}_i^H)^T \tilde{\sigma}_{n+1}^H d\xi d\eta d\zeta \quad (55)$$

In particular, the hourglass force needs to be decomposed into different linear term integral forms as:

$$\begin{aligned} \mathbf{F}_i^{stab} = & \int_{-1}^1 \int_{-1}^1 \int_{-1}^1 (\tilde{\mathbf{B}}_{1i}^H)^T \tilde{\sigma}_1^{(n+1)H} d\xi d\eta d\zeta \\ & + \int_{-1}^1 \int_{-1}^1 \int_{-1}^1 (\tilde{\mathbf{B}}_{2i}^H)^T \tilde{\sigma}_2^{(n+1)H} d\xi d\eta d\zeta \\ & + \int_{-1}^1 \int_{-1}^1 \int_{-1}^1 (\tilde{\mathbf{B}}_{4i}^H)^T \tilde{\sigma}_4^{(n+1)H} d\xi d\eta d\zeta \\ & + \int_{-1}^1 \int_{-1}^1 \int_{-1}^1 (\tilde{\mathbf{B}}_{5i}^H)^T \tilde{\sigma}_5^{(n+1)H} d\xi d\eta d\zeta \end{aligned} \quad (56)$$

4. Numerical tests

In this section, the ability of the element to eliminate the locking phenomenon and the computational efficiency are verified, respectively. The performance of the above element can be demonstrated through a series of verification examples. Cookmembrane and roof shells were tested for shear locking, and membrane locking, the hemispherical shell as a complex condition test should be coupled to the

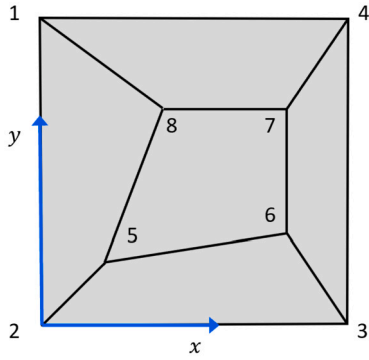


Fig. 5. The plate bending patch test.

Table 1

Nodal coordinates and exact displacements and rotations for the bending patch test.

Node	dof				
	x	y	w	θ_x	θ_y
1	0.0	10.0	0.54	0.06	-0.045
2	0.0	0.0	0.09	0.03	-0.025
3	10.0	0.0	0.44	0.05	-0.045
4	10.0	10.0	1.09	0.08	-0.065
5	2.0	2.0	0.2180	0.04	-0.033
6	8.0	3.0	0.5055	0.055	0-0.047
7	8.0	7.0	0.7495	0.067	-0.055
8	4.0	7.0	0.5455	0.058	-0.047

strain of both the membrane and the bending component, and the convergence is evaluated using a flat plate. Finally, A series of arithmetic examples were then used to test the computational efficiency in explicit dynamic scheme. The MITC4+ element was chosen as the reference for comparison.

4.1. Static problems

4.1.1. The plate bending patch test

The interesting plate bending patch test is introduced in this work. The mesh for the plate is given in Fig. 5. The nodal coordinates are shown in the table, where the w field can be calculated by the following

$$w(x, y) = 0.001x^2 + 0.002xy + 0.0015y^2 + 0.025x + 0.03y + 0.09 \quad (57)$$

The coefficients of the above polynomial are arbitrarily chosen. Under the Kirchhoff-Love assumption, the rotation can be calculated as follows:

$$\begin{aligned} \theta_x(x, y) &= \frac{\partial w}{\partial y} = 0.002x + 0.003y + 0.003 \\ \theta_y(x, y) &= -\frac{\partial w}{\partial x} = -(0.002x + 0.002y + 0.025) \end{aligned} \quad (58)$$

The exterior nodes 1-4 are loaded with exact transverse displacement and rotations, and the interior nodes are observed to match the analytic solution has been showed in Table 1. The plate bending patch test verifies the ability of the element to detect the convergence of the shell. The results of MITC4+ and MITC4+R calculations keep the perfect agreement with the analytical solution.

4.1.2. Zero-energy mode test

Zero-energy mode test is widely used to test whether an element can correctly represent a rigid body mode. The number of non-zero eigenvalues for different mesh cells is considered as shown in Fig. 6. Young's modulus $E = 1.5 \times 10^3$, Poisson's ratio $\nu = 0.3$ and thickness $t = 1.0$.

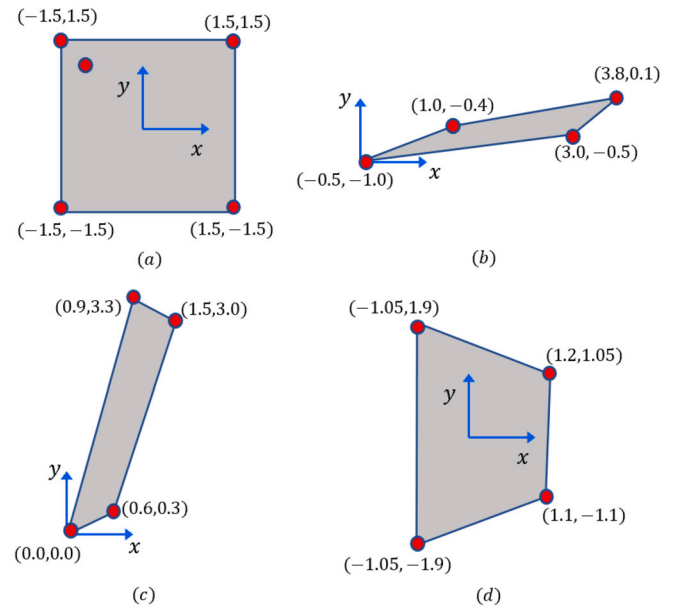


Fig. 6. Zero-energy mode test.

Table 2

Results for Cookmembrane.

Element	Mesh		
	4x4	8x8	16x16
MITC4+R	1.8836e+01	2.2732e+01	2.4403e+01
MITC4+	1.8580e+01	2.2665e+01	2.4403e+01

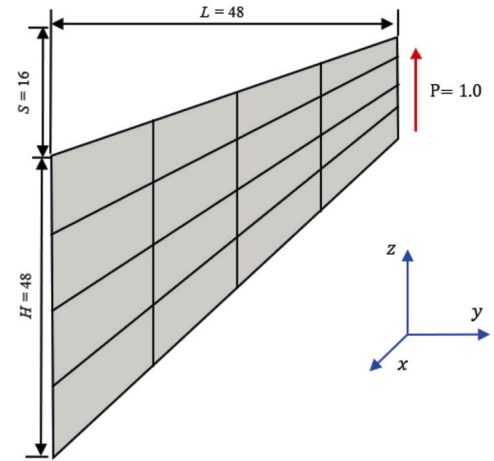


Fig. 7. Cookmembrane.

Table 3

Results for Scordelis-Lo roof.

Element	Mesh		
	4x4	8x8	16x16
MITC4+R	2.8316e-01	2.9616e-01	3.0672e-01
MITC4+	2.8316e-01	2.9616e-01	3.0671e-01

The MITC4+R contains only six zero eigenvalues and produces six rigid body modes (three translational and three rotational modes) correctly in all geometric, without any spurious energy modes, arbitrary to the deformation of the elements. Hence this element passed the zero energy mode test.

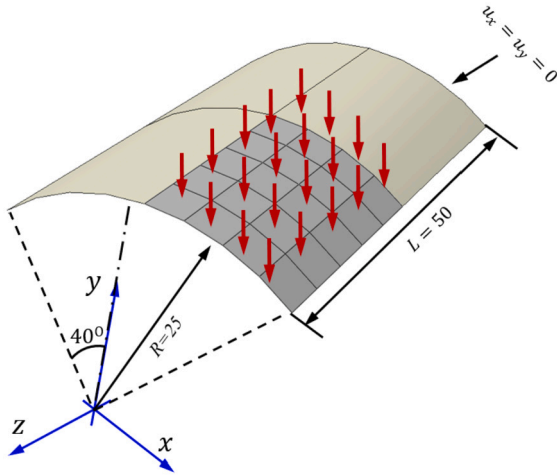


Fig. 8. The Scordelis-Lo roof.

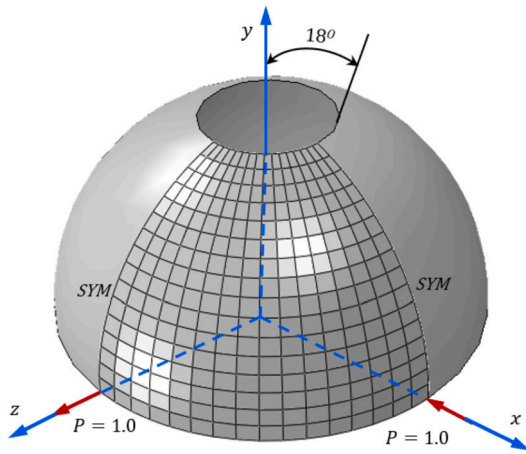


Fig. 9. The Spherical shell.

4.1.3. Cookmembrane

The Cookmembrane problem is considered. The famous standard example for in-plane strain analysis is shown in Fig. 7. Young's modulus $E = 1.0$, Poisson's ratio $\nu = 0.33$ and thickness $t = 1.0$. One edge is clamped, and a unit shear force is applied to the other. The reference solution for the vertical displacement at the midpoint of the force edge is 23.91. The results of the MITC4+ and MITC4+R are listed in Table 2. Both of them show better in-plane behaviour.

4.1.4. Scordelis-Lo roof

The Scordelis-Lo roof is a classic example involving both membrane and bending deformation. It is used as a standard in many commercial applications such as Altair and COMSOL. The Scordelis-Lo roof is the singly-curved shell problem, and the quarter model is considered to gravity load shown in Fig. 8, where Young's modulus $E = 4.32 \times 10^8$, Poisson's ratio $\nu = 0.0$ and thickness $t = 0.25$, the vertical displacement of the midpoint of the free edge is mainly observed; the theoretical value of the point is 0.3086. The results of the MITC4+ and MITC4+R are listed in Table 3. The same accuracy is shown in the results of each.

4.1.5. Spherical shell

A hemispherical shell is a double curvature problem, a relatively complex example of studying membrane and bending strain. There are two models of the hemispherical shell with holes and without holes. The spherical shell with holes can effectively avoid dividing triangular elements. The geometric model is shown in Fig. 9, where $R = 10$, Young's modulus $E = 6.852 \times 10^7$, Poisson's ratio $\nu = 0.3$ and thickness $t = 0.04$,

Table 4
Results for spherical shell.

Element	Mesh		
	4x4	8x8	16x16
MITC4+R	5.8139e-02	9.0627e-02	9.2000e-02
MITC4+	5.9359e-02	9.0197e-02	8.8032e-02

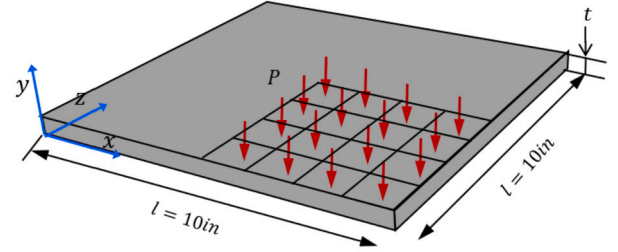


Fig. 10. The clamped square plate problem.

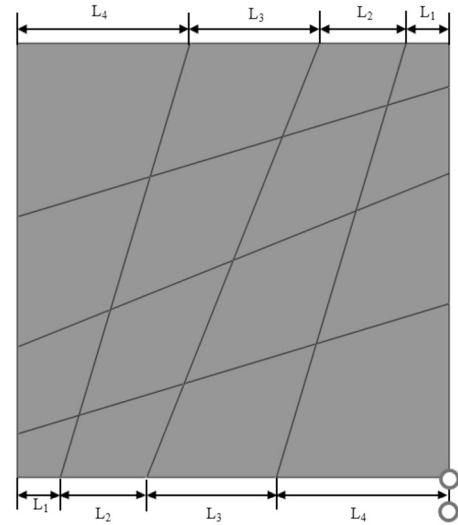


Fig. 11. The clamped square plate problem for distorted mesh.

a quarter of model is considered. The bottom is the free edge to apply a radial load every 90 degrees. The displacement reference solution at the loading point is 0.9024. The results of the MITC4+ and MITC4+R are shown in Table 4. It can be seen that MITC4+R has shown better performance.

4.1.6. Convergence studies

Consistency and stability are two important requirements for finite element convergence. An appropriate norm can well detect the convergence consistency and convergence rate of shell elements. The s-norm method proposed by Hiller and Bathe [39] measure the error of shell problem.

$$E_h = \frac{\|\mathbf{u}_{ref} - \mathbf{u}_h\|_s^2}{\|\mathbf{u}_{ref}\|_s^2} \quad (59)$$

$$\|\mathbf{u} - \mathbf{u}_h\|_s^2 = \int_{\Omega} \Delta \epsilon^T \Delta \tau d\Omega \quad \text{with } \Delta \epsilon = \epsilon - \epsilon_h, \Delta \tau = \tau - \tau_h, \quad (60)$$

where $\mathbf{u}$ is the analytical solution, and ϵ and τ are the strain and stress vectors. Since there is no specific analytical solution for some shell examples, Lee et al. [40]. $\mathbf{u}$ was replaced with the exact solution $\mathbf{u}_{ref}$ for the very fine mesh calculation.

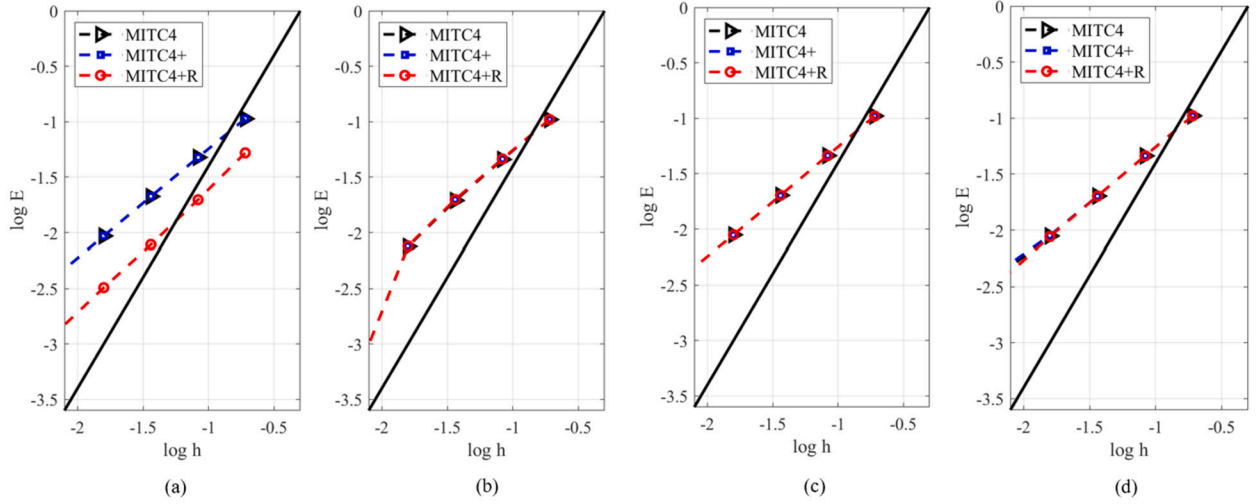


Fig. 12. Convergence curves for the fully clamped square plate problem with: (a) $h/l = 100$ (b) $h/l = 1000$ (c) $h/l = 10000$.

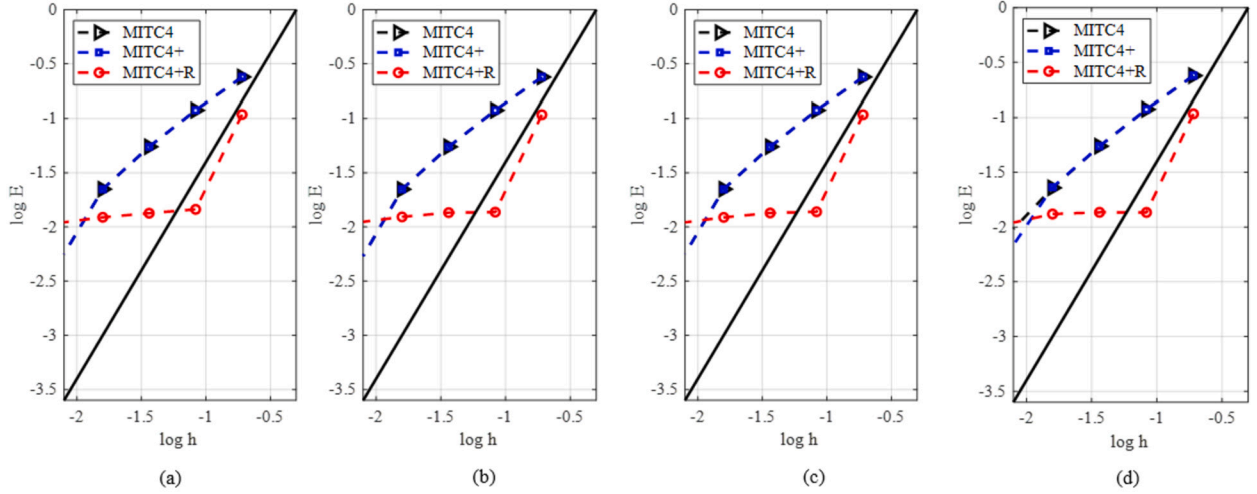


Fig. 13. Convergence curves for the fully clamped square plate problem for distorted mesh: (a) $h/l = 100$ (b) $h/l = 1000$ (c) $h/l = 10000$.

$$\|\mathbf{u}_{ref} - \mathbf{u}_h\|_s^2 = \int_{\Omega_{ref}} \Delta \epsilon^T \Delta \tau d\Omega_{ref} \quad \text{with } \Delta \epsilon = \epsilon_{ref} - \epsilon_h, \Delta \tau = \tau_{ref} - \tau_h. \quad (61)$$

The normalization error E_h to study the effect of shell thickness on the solution convergence. The clamped plate problem was used to measure the convergence of the element in Fig. 10 and the distorted mesh is considered as shown in Fig. 11. Each edge is decomposed in the following scale: $L_1 : L_2 : L_3 : \dots : L_n = 1 : 2 : 3 : \dots : N$. A square plate subjected to uniform pressure, a quarter of the model is considered, where Young's modulus $E = 1.7472 \times 10^7$, Poisson's ratio $\nu = 0.3$ and three cases of plate thickness of 0.1, 0.01, 0.001, and 0.0001 were examined separately. A 72×72 mesh of S8R in Abaqus is obtained the reference solution. Its convergence is shown in Fig. 12. It can be seen that the MITC4+ and MITC4+R convergence remain similar almost under the regular mesh except for $1/h=10$, but there is a slight difference between the two when the distorted mesh is used. (See Fig. 13.)

4.2. Dynamic problems

Automatic time increments are used in explicit dynamics problem. The increment of time can be defined by the element length and the wave speed of the material as follows:

$$\Delta t_{stable} = \frac{L^e}{C_d} \quad (62)$$

L^e denotes the shortest length of the element, while the wave velocity C_d is a property of the material. This algorithm is used for all explicit dynamics problems in the latter sections and the computations are performed using C++ codes on an Intel Core i5-10400F @ 2.90 GHz, 16-GB RAM, 64-bit Windows 10 operating system.

4.2.1. Shallow spherical dome

The shallow spherical shell dome is excellent for examining the convergence properties of the shell elements under uniform pressure. Due to the symmetric model, only a quarter part is considered. Geometric parameters and boundary conditions are shown in Fig. 14 Fully elastomeric material with Young's modulus $E = 1.05 \times 10^7$ psi, Poisson's ratio $\nu = 0.3$, density $\rho = 2.45 \times 10^{-4}$ psi s²/in², initial yield stress $\sigma_0 = 2.4 \times 10^4$ psi, and plastic modulus $H = 2.4 \times 10^5$ psi was used.

The vertical deflection of two points as a function of time is considered, one for the whole model shallow spherical shell (Point A) and one for the quarter model (Point B). The results are shown in Fig. 15 and Fig. 16, respectively. It can be seen that the numerical solutions of MITC4+ and MITC4+R are in good agreement with the S4R of Abaqus [41].

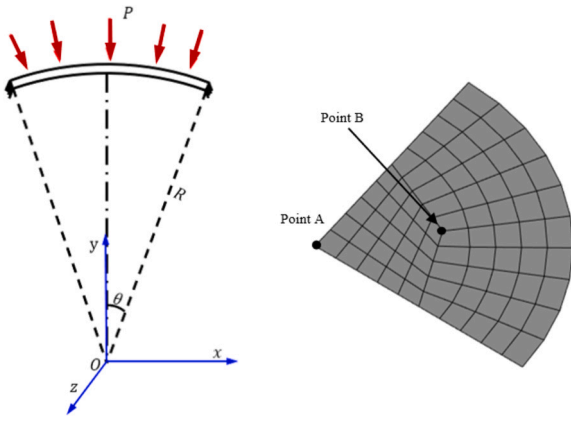


Fig. 14. The shallow spherical.

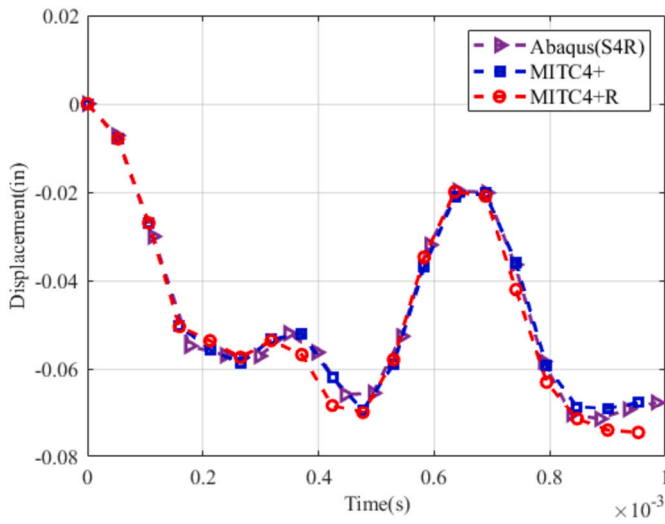


Fig. 15. Dynamic response of Shallow spherical dome at the point A.

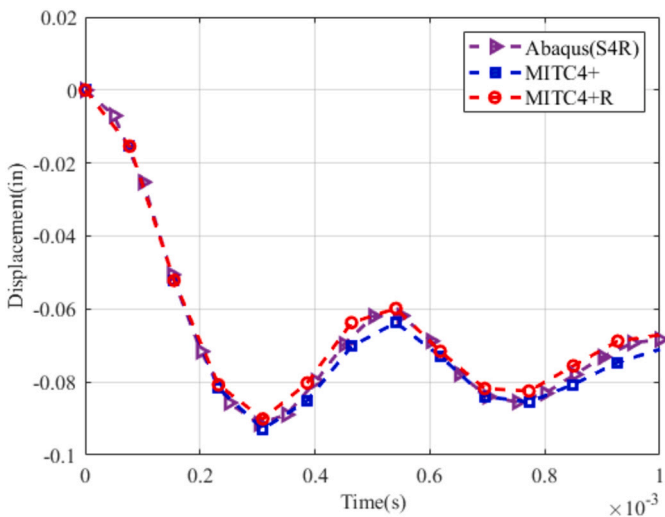


Fig. 16. Dynamic response of Shallow spherical dome at point B.

4.2.2. Twist beam

The twisted beam is a classic example to test the warping problem. The beam is twisted by 90-degree, and when the division is coarse mesh (2×12), each element is approximately 7.5-degree warped. Geometric parameters is seen in Fig. 17, where Young's modulus $E = 2.9 \times 10^7$,

Poisson's ratio $\nu = 0.22$, thickness $t = 0.32$ and density $\rho = 2.588 \times 10^{-4}$ psi s^2/in^2 . Both out-of-plane and in-plane load are considered.

The dynamic responses of MITC4+R and MITC4+ elements are plotted in Fig. 18 and Fig. 19, respectively. The expected result are 0.005424 and 0.001754 for the dynamic solution. Both MITC4+ and MITC4+R are in general agreement with the results of Abaqus' S4R.

4.2.3. Cylinder panel

The cylinder panel is being tested as a large deformation elastic-plastic material model, a thin-walled cylindrical shell using a fully elastic-plastic model. Both ends are clamped, and an initial velocity of 5650 in/s is applied in the middle region see in Fig. 20. Young's modulus $E = 1.05 \times 10^7$ psi, Poisson's ratio $\nu = 0.33$, density $\rho = 2.5 \times 10^{-4}$ psi s^2/in^2 , initial yield stress $\sigma_0 = 4.4 \times 10^4$ psi, and plastic modulus $H = 0$ psi was used. Half of the model is analysed to verify the dynamic response at the centre (6.28 in) of the symmetry edge and three-quarters (9.42 in) of the span.

Deformed shapes at different time are shown in Fig. 21. Fig. 22 and Fig. 23 show the deformation process at 6.28 in and 9.42 in of the cylinder panel, respectively. It has been observed that the results for both MITC4+ and MITC4+R are consistent with S4R at the point $z = 6.28$ in, and the results are close relative to the experiments done by ALuigi et al. [42], but the results of all of them are still different from the experimental solution at $z = 9.42$ in.

4.2.4. Impact between two tubes

The geometric model of the two thin tubes is shown in Fig. 24. The material is considered elastic-plastic with large strain. Young's modulus $E = 2 \times 10^{11}$ psi, Poisson's ratio $\nu = 0.3$, density $\rho = 9840 \text{ kg/m}^3$, initial yield stress $\sigma_0 = 2 \times 10^8$ psi, and plastic modulus $H = 0$ psi was used. The initial velocity of the left tube is (80, 0, 0) m/s, and the right is (-80, 0, 30) m/s. MITC4+ and MITC4+R were used to solve this problem. Some of the deformation processes are shown in Fig. 25. The results evaluated that MITC4+R has a great ability to handle large deformations in contact.

4.2.5. A ball impacting a clamped beam

In this work, a ball impacts a claimed beam as shown in Fig. 26. The initial velocity of the ball is 30 m/s. Both ball and plate are elastic material, Young's modulus $E = 2 \times 10^{11}$ psi, Poisson's ratio $\nu = 0.3$, density $\rho = 9800 \text{ kg/m}^3$, MITC4+ and MITC4+R were used to solve this problem. Fig. 27 shows the deformation of the system at (a) 2 ms, (b) 4 ms and (c) 8 ms, respectively. Their deformation schema is consistent with the results of Cardoso et al. [35].

4.3. Comparison of calculation time

Finally, the computational efficiency of MITC4+ and MITC4+R is compared. For a better view of the results, the above five examples of explicit dynamics are examined, the calculation time of MITC4+ and MITC4+R was counted for each example. The final results are shown in Table 5. All computation times in the table are in seconds, and the letters in the second row of the table indicate the different tests: (a) shallow spherical dome, (b) twist beam, (c) cylindrical panel, (d) impact between two tubes, and (e) ball impact clamped beam, respectively. It can be clearly observed that MITC4+R has a higher computational efficiency in the previous four examples. Due to the division of the sphere ball into a large number of solid elements, only a weak advantage is shown in the last example. In conclusion, the MITC4+R has a higher computational rate than the MITC4+ at the same computational scale.

5. Conclusions

In this work, we derive a novel reduced integral quadrilateral shell element based on the MITC4+ method. In the patch test, all the computational results data of this element match the analytical solution. The

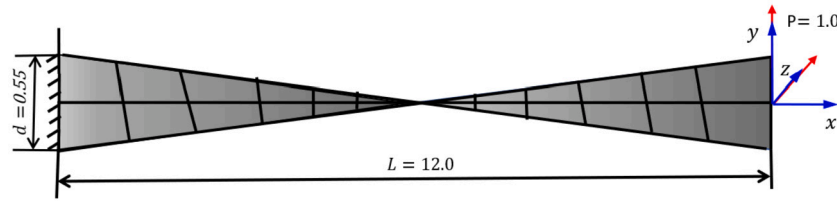


Fig. 17. The twist beam.

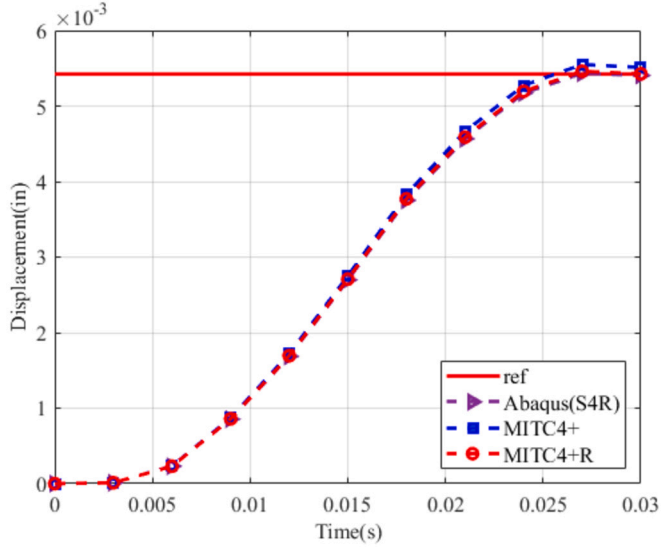


Fig. 18. Comparison of the dynamic response for the twist beam (in-plane).

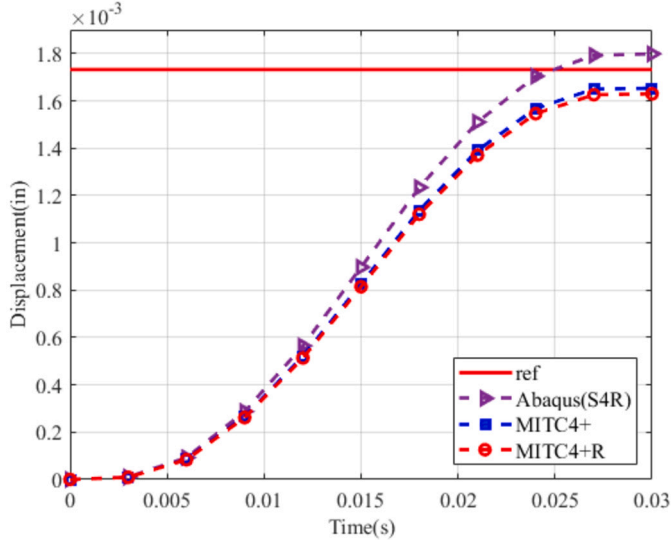


Fig. 19. Comparison of the dynamic response for the twist beam (out-of-plane).

Cookmembrane problem was tested, and the behaviour of the in-plane membrane strain was simulated very successfully. According to the tests with Scordelis-Lo roof, it was also proven that the proposed formulation provides almost unique results in membrane and bending deformation. Based on hemispherical shell testing, the developed shell provides excellent alleviation of membrane locking when elements are warped. Then, the convergence studies verified that the shell kept a great agreement with MITC4+ convergence. Finally, a series of explicit dynamics tests, such as Shallow spherical dome, Cylinder panel, and Twist beam illustrate that the element exhibits higher computational efficiency without

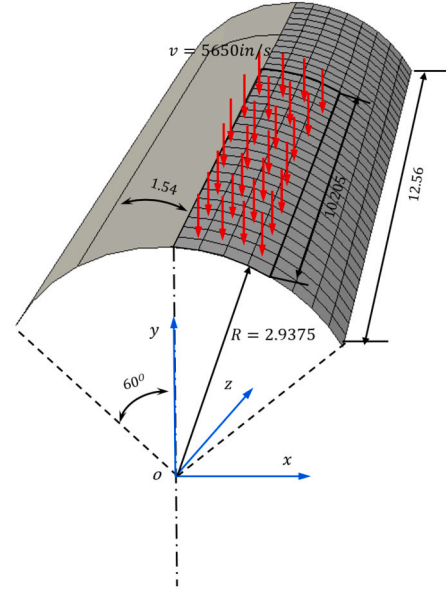


Fig. 20. The Cylinder panel under impulse load.

loss of MITC4+ accuracy. Enables the present MITC4+R element to be more potentially applied to engineering applications.

Declaration of competing interest

The authors declared no potential conflicts of interest with respect to the research, author-ship, or publication of this article.

Data availability

No data was used for the research described in the article.

Acknowledgements

The support of National Science Foundation of China (12002124), Hunan Provincial Natural Science Foundation (2022JJ40031), National Key Research and Development Program of China (2022YFB2503505), China Postdoctoral Science Foundation (2021M690965), Changsha Municipal Natural Science Foundation (kq2014035), and State Key Laboratory of Advanced Design and Manufacturing for Vehicle Body (32065007) are gratefully acknowledged.

Appendix A. B matrix of transverse shear and bending component

In this section, the B-matrices of the transverse shear and bending components are derived in detail. Eq. (34), Eq. (35) and Eq. (37) can be rewritten for convenience as

$$\mathbf{J}^m = \begin{bmatrix} \frac{\partial \mathbf{x}_m}{\partial \eta} \\ \frac{\partial \mathbf{x}_m}{\partial \xi} \end{bmatrix} = \frac{1}{4} \begin{bmatrix} J_{11}^{m0} & J_{12}^{m0} & J_{13}^{m0} \\ J_{21}^{m0} & J_{22}^{m0} & J_{23}^{m0} \\ 0 & 0 & 0 \end{bmatrix} + \frac{1}{4} \begin{bmatrix} J_{11}^{m1} \eta & J_{12}^{m1} \eta & J_{13}^{m1} \eta \\ J_{21}^{m1} \xi & J_{22}^{m1} \xi & J_{23}^{m1} \xi \\ 0 & 0 & 0 \end{bmatrix}, \quad (\text{A.1})$$

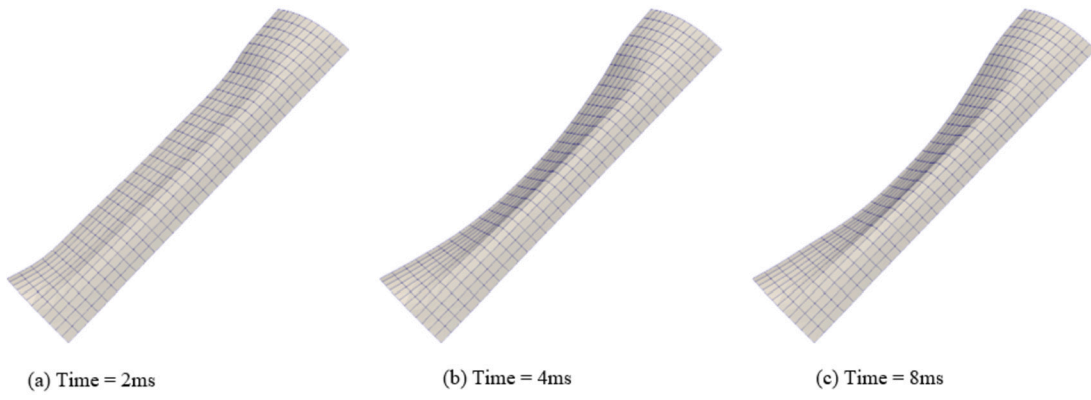


Fig. 21. Deformation process of cylinder panel.

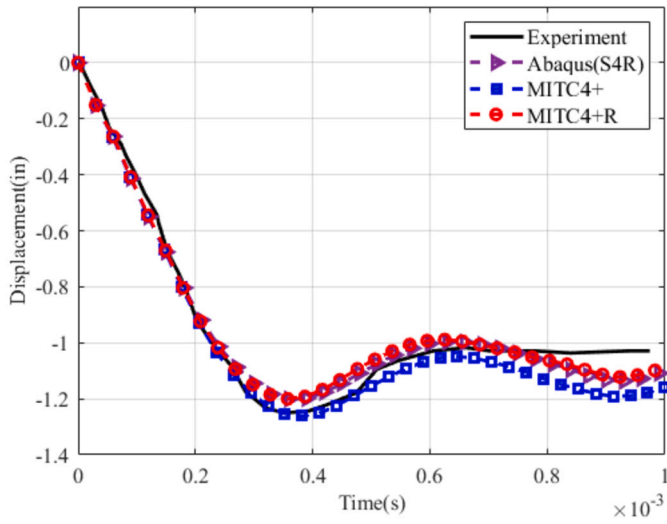


Fig. 22. Dynamic response of the cylinder panel along the symmetry line at point $z = 6.28$ in.

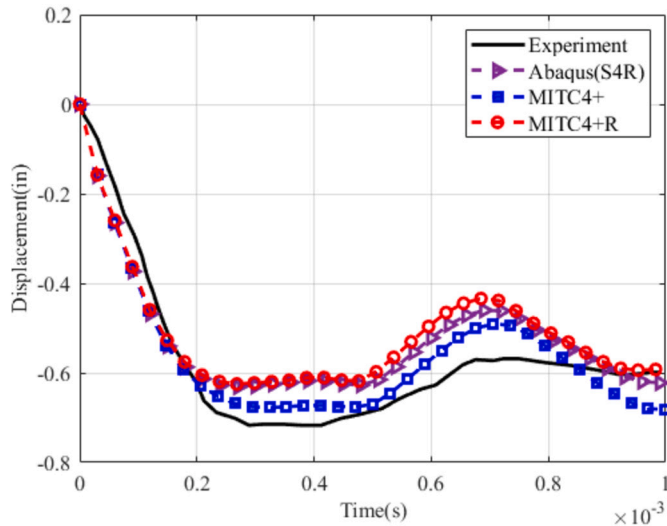


Fig. 23. Dynamic response of the cylinder panel along the symmetry line at point $z = 9.42$ in.

$$\mathbf{J}^b = \begin{bmatrix} \frac{\partial \mathbf{x}_b}{\partial \eta} \\ \frac{\partial \mathbf{x}_b}{\partial \xi} \\ \frac{\partial \mathbf{x}_b}{\partial \zeta} \end{bmatrix} = \frac{a}{8} \begin{bmatrix} J_{11}^{b0} & J_{12}^{b0} & J_{13}^{b0} \\ J_{21}^{b0} & J_{22}^{b0} & J_{23}^{b0} \\ 0 & 0 & 0 \end{bmatrix} + \frac{a}{8} \begin{bmatrix} J_{11}^{b1} \eta & J_{12}^{b1} \eta & J_{13}^{b1} \eta \\ J_{21}^{b1} \xi & J_{22}^{b1} \xi & J_{23}^{b1} \xi \\ 0 & 0 & 0 \end{bmatrix} \quad (\text{A.2})$$

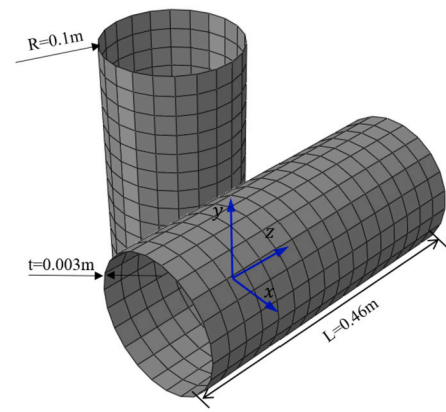


Fig. 24. Deformation process of tube impact system.

and

$$\mathbf{J} = \begin{bmatrix} \frac{\partial \mathbf{x}}{\partial \eta} \\ \frac{\partial \mathbf{x}}{\partial \xi} \\ \frac{\partial \mathbf{x}}{\partial \zeta} \end{bmatrix} = \frac{1}{4} \begin{bmatrix} J_{11}^0 & J_{12}^0 & J_{13}^0 \\ J_{21}^0 & J_{22}^0 & J_{23}^0 \\ J_{31}^0 & J_{32}^0 & J_{33}^0 \end{bmatrix} + \frac{1}{4} \begin{bmatrix} J_{11}^1 \eta & J_{12}^1 \eta & J_{13}^1 \eta \\ J_{21}^1 \xi & J_{22}^1 \xi & J_{23}^1 \xi \\ 0 & 0 & 0 \end{bmatrix} \quad (\text{A.3})$$

The detailed expression of the transverse shear and bending strain component $\mathbf{B}$ is as follows:

$$\bar{\mathbf{B}}_{1i}^{bs0} = \frac{1}{4} \begin{bmatrix} 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{a_i}{2} B_{13i}^0 & -\frac{a_i}{2} B_{23i}^0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0.5\phi_{1ACi}^0 & 0.5\phi_{2ACi}^0 & 0.5\phi_{3ACi}^0 & -\frac{a_i}{4} \psi_{1ACi}^0 & -\frac{a_i}{4} \psi_{2ACi}^0 \\ 0.5\phi_{1DBi}^0 & 0.5\phi_{2DBi}^0 & 0.5\phi_{3DBi}^0 & -\frac{a_i}{4} \psi_{1DBi}^0 & -\frac{a_i}{4} \psi_{2DBi}^0 \end{bmatrix}, \quad (\text{A.4})$$

$$\bar{\mathbf{B}}_{2i}^{bs0} = \frac{1}{4} \begin{bmatrix} \xi_i J_{11}^{b0} & \xi_i J_{12}^{b0} & \xi_i J_{13}^{b0} & -\frac{a_i}{2} \xi_i B_{12i}^{m0i} & \frac{a_i}{2} \xi_i B_{11i}^{m0i} \\ \eta_i J_{21}^{b0} & \eta_i J_{22}^{b0} & \eta_i J_{23}^{b0} & -\frac{a_i}{2} \eta_i B_{20i}^{m0i} & \frac{a_i}{2} \eta_i B_{11i}^{m0i} \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix} \quad (\text{A.5})$$

$$+ \frac{1}{4} \begin{bmatrix} 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ \eta_i \xi_i J_{21}^{b0} & \eta_i \xi_i J_{22}^{b0} & \eta_i \xi_i J_{23}^{b0} & -\frac{a_i}{2} \xi_i B_{22i}^{m0i} & \frac{a_i}{2} \xi_i B_{21i}^{m0i} \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}.$$

Where, the corrected hourglass terms of the transverse shear bending component become

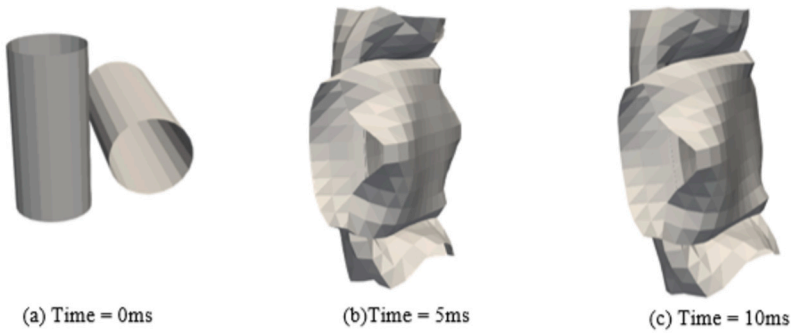


Fig. 25. Deformation process of tube impact system.

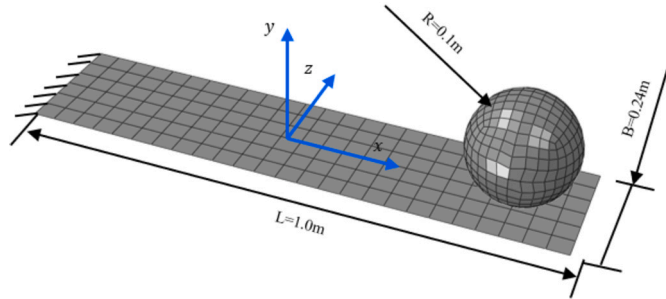


Fig. 26. Deformation process of two ball.

Table 5

Comparison of computational efficiency.

Element	test				
	a	b	c	d	e
MITC4+R	1.73	14.83	4.76	11.08	349.67
MITC4+	4.84	28.63	15.84	59.5	698.96

$$\overline{\mathbf{B}}_{1i}^{bsH} = \frac{1}{4} \begin{bmatrix} 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{a_i}{2} \xi_i B_{13}^{0i} & -\frac{a_i}{2} \xi_i B_{23}^{0i} \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0.5\phi_{1DBi}^H & 0.5\phi_{2DBi}^H & 0.5\phi_{3DBi}^H & -\frac{a_i}{4} \psi_{1DBi}^H & -\frac{a_i}{4} \psi_{2DBi}^H \end{bmatrix}, \quad (\text{A.6})$$

$$\overline{\mathbf{B}}_{2i}^{bsH} = \frac{1}{4} \begin{bmatrix} 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{a_i}{2} \eta_i B_{13}^{0i} & -\frac{a_i}{2} \eta_i B_{23}^{0i} \\ 0 & 0 & 0 & 0 & 0 \\ 0.5\phi_{1ACi}^H & 0.5\phi_{2ACi}^H & 0.5\phi_{3ACi}^H & -\frac{a_i}{4} \psi_{1ACi}^H & -\frac{a_i}{4} \psi_{2ACi}^H \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}, \quad (\text{A.7})$$

$$\overline{\mathbf{B}}_{4i}^{bsH} = \frac{1}{4} \begin{bmatrix} 0 & 0 & 0 & 0 & 0 \\ h_i J_{21}^{b0} & h_i J_{22}^{b0} & h_i J_{23}^{b0} & -\frac{a_i}{2} h_i B_{22}^{m0i} & \frac{a_i}{2} h_i B_{21}^{m0i} \\ 0 & 0 & 0 & 0 & 0 \\ h_i J_{11}^{b0} & h_i J_{12}^{b0} & h_i J_{13}^{b0} & -\frac{a_i}{2} h_i B_{12}^{m0i} & \frac{a_i}{2} h_i B_{11}^{m0i} \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix} + \frac{1}{4} \begin{bmatrix} 0 & 0 & 0 & 0 & 0 \\ \eta_i J_{21}^{b1} & \eta_i J_{22}^{b1} & \eta_i J_{23}^{b1} & -\frac{a_i}{2} \eta_i B_{22}^{m1i} & \frac{a_i}{2} \eta_i B_{21}^{m1i} \\ 0 & 0 & 0 & 0 & 0 \\ \xi_i J_{21}^{b1} & \xi_i J_{22}^{b1} & \xi_i J_{23}^{b1} & -\frac{a_i}{2} \xi_i B_{22}^{m1i} & \frac{a_i}{2} \xi_i B_{21}^{m1i} \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}, \quad (\text{A.8})$$

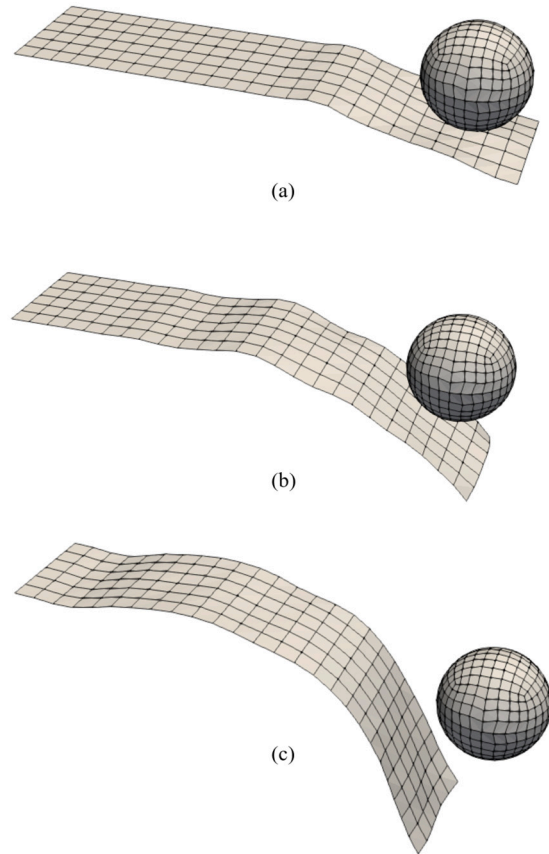


Fig. 27. Deformation process of two ball.

$$\overline{\mathbf{B}}_{5i}^{bsH} = \frac{1}{4} \begin{bmatrix} h_i J_{11}^{b0} & h_i J_{12}^{b0} & h_i J_{13}^{b0} & -\frac{a_i}{2} h_i B_{12}^{m0i} & \frac{a_i}{2} h_i B_{11}^{m0i} \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ h_i J_{21}^{b0} & h_i J_{22}^{b0} & h_i J_{23}^{b0} & -\frac{a_i}{2} h_i B_{22}^{m0i} & \frac{a_i}{2} h_i B_{21}^{m0i} \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix} + \frac{1}{4} \begin{bmatrix} \xi_i J_{21}^{b1} & \xi_i J_{22}^{b1} & \xi_i J_{23}^{b1} & -\frac{a_i}{2} \xi_i B_{22}^{m1i} & \frac{a_i}{2} \xi_i B_{21}^{m1i} \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ \xi_i J_{11}^{b1} & \xi_i J_{12}^{b1} & \xi_i J_{13}^{b1} & -\frac{a_i}{2} \xi_i B_{12}^{m1i} & \frac{a_i}{2} \xi_i B_{11}^{m1i} \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}. \quad (\text{A.9})$$

The term of $B_{\alpha\beta}^{ik}$ are given by

$$B_{\alpha\beta}^{ik} = \mathbf{J}_{\alpha}^i \cdot \mathbf{V}_{\beta}^k, \quad (\text{A.10})$$

where $\alpha = 1 - 3$; $\beta = 1 - 2$; $i = 0, m0, m1$, the transverse shear interpolation coefficients by

$$\begin{aligned} \phi_{\alpha ACi}^0 &= J_{3x}^A (\xi_i + h_i) + J_{3x}^C (\xi_i - h_i) & \phi_{\alpha DBi}^0 &= J_{3x}^D (\eta_i + h_i) + J_{3x}^B (\eta_i - h_i), \\ \psi_{\beta ACi}^0 &= B_{\beta 1i}^A (1 + \eta_i) + B_{\beta 1i}^C (1 - \eta_i) & \psi_{\beta DBi}^0 &= B_{\beta 2i}^D (1 + \xi_i) + B_{\beta 2i}^B (1 - \xi_i), \\ \phi_{\alpha ACi}^H &= J_{3x}^A (\xi_i + h_i) - J_{3x}^C (\xi_i - h_i) & \phi_{\alpha DBi}^H &= J_{3x}^D (\eta_i + h_i) - J_{3x}^B (\eta_i - h_i), \\ \psi_{\beta ACi}^H &= B_{\beta 1i}^A (1 + \eta_i) - B_{\beta 1i}^C (1 - \eta_i) & \psi_{\beta DBi}^H &= B_{\beta 2i}^D (1 + \xi_i) - B_{\beta 2i}^B (1 - \xi_i) \end{aligned} \quad (A.11)$$

where $J_{\alpha}^{A,B,C,D}$ is the Jacobian terms in sampling point.

$$B_{\alpha\beta i}^{A,B,C,D} = J_{\alpha}^{A,B,C,D} \cdot V_{\beta}^i, \quad (A.12)$$

and $\alpha = 1 - 3$; $\beta = 1 - 2$; $i = 1 - 4$.

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